

The cylinder — surface area and volume

A prism whose base is a circle: the same two formulas, with π in them.

LESSON 28–29

2 × 40 MIN

GEOMETRY 11, §14

IGCSE E13.X

LESSON CLOCK

12'

22'

22'

22'

20'

8'

A prism with a circular base	12 min
The net and the area	22 min
The volume	22 min
Working backwards	22 min
Compound solids	20 min
Homework	8 min

LEARNING OBJECTIVES

- Derive the surface area of a cylinder from its net.
- Compute the volume of a cylinder.
- Solve for a missing radius or height.
- Handle hollow cylinders and compound solids.

TEXTBOOK

National	pp. 141–152
Cambridge	Core & Extended, pp. 311–318
Workbook	Exercise 14.1

01

Explanation

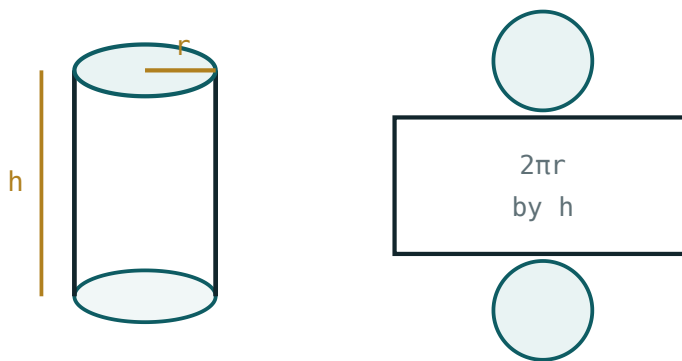
A prism with a circular base

A **cylinder** is the solid swept out when a rectangle turns about one of its sides. Its two bases are congruent circles in parallel planes, and every cross-section parallel to the base is the same circle.

So the principle of the last lesson applies unchanged: **uniform cross-section means $V = Bh$** , with $B = \pi r^2$.

$$V = \pi r^2 h$$

The net and the surface area



$$S = 2\pi r^2 + 2\pi rh - \text{two circles and one rectangle}$$

Cut down one side and unroll: two circles and a rectangle $2\pi r$ wide.

The curved surface unrolls into a rectangle whose width is the circumference of the base and whose height is the cylinder's height:

$$S_{\text{curved}} = 2\pi rh \quad S_{\text{total}} = 2\pi r^2 + 2\pi rh = 2\pi r(r + h)$$

CURVED, OR TOTAL?

A tin has two ends. A pipe has none. A cup has one. Read the object before choosing the formula — the question always describes it precisely enough to decide.

Working backwards

Given a volume and one dimension, the other follows:

$$r = \sqrt{\frac{V}{\pi h}} \quad h = \frac{V}{\pi r^2}$$

Example. A cylindrical tin holds 500 ml and is 12 cm tall. Its radius is $\sqrt{\frac{500}{12\pi}} \approx 3.64$ cm.

Given a **surface area** and one dimension, the equation is quadratic in r — solve it with the formula and reject the negative root.

Hollow and compound solids

A pipe of outer radius R and inner radius r is the difference of two cylinders:

$$V = \pi(R^2 - r^2)h$$

Do **not** compute $\pi(R - r)^2h$ — the cross-section is an annulus, whose area is a difference of two circles, not the circle on the difference of the radii.

Compound solids. A silo is a cylinder with a hemisphere on top; a pencil is a hexagonal prism with a cone. Split, compute each, add — and remember that the joining faces are internal and contribute nothing to the surface area.

THE AXIAL SECTION

Cutting a cylinder through its axis gives a rectangle $2r$ by h . A cylinder whose axial section is a square has $h = 2r$ — the shape that minimises the surface area for a fixed volume, as Lesson 23–25 of Quarter I showed.

02 Worked examples

EXAMPLE 1 A cylinder has radius 5 cm and height 12 cm. Find its volume and total surface area.

$$\begin{aligned} 1 \quad V &= \pi \times 25 \times 12 = 300\pi \\ &\approx 942.5 \text{ cm}^3 \end{aligned}$$

$$2 \quad S = 2\pi(25) + 2\pi(5)(12)$$

$$\begin{aligned} 3 \quad &= 50\pi + 120\pi = 170\pi \\ &\approx 534.1 \text{ cm}^2 \end{aligned}$$

ANSWER

$$300\pi \approx 942 \text{ cm}^3; 170\pi \approx 534 \text{ cm}^2$$

EXAMPLE 2 A cylindrical tin holds 500 ml and is 12 cm tall. Find its radius.

$$\textcircled{1} \quad 500 = \pi r^2 \times 12$$

$$1 \text{ ml} = 1 \text{ cm}^3.$$

$$\textcircled{2} \quad r^2 = \frac{500}{12\pi} \approx 13.26$$

$$\textcircled{3} \quad r \approx 3.64$$

ANSWER

$\approx 3.64 \text{ cm}$

EXAMPLE 3 A pipe is 2 m long with outer radius 8 cm and inner radius 6 cm. Find the volume of metal.

$$\textcircled{1} \quad V = \pi(64 - 36) \times 200$$

$$\text{Length in cm.}$$

$$\textcircled{2} \quad = \pi \times 28 \times 200$$

$$\textcircled{3} \quad = 5600\pi \approx 17\,593$$

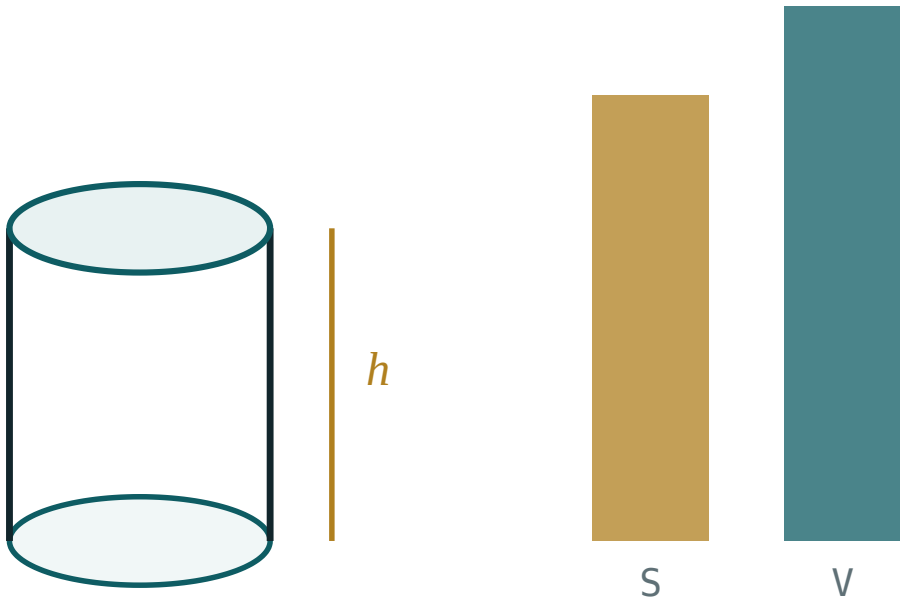
ANSWER

$5600\pi \approx 17\,600 \text{ cm}^3$

03 Interactive model

Unroll a tin label and measure the rectangle — its width is the circumference.

The cylinder



base area B **50.3**

perimeter P **25.1**

total S **251.3**

volume V **301.6**

04 Terminology

Write these on the board at the start. Learners meet the national programme in Uzbek or Russian and the Cambridge papers in English — the three columns are the same idea.

ENGLISH	O'ZBEKCHA	РУССКИЙ
Cylinder	Silindr	Цилиндр
Axis of a cylinder	Silindr o'qi	Ось цилиндра
Base circle	Asos aylanasi	Основание
Curved surface	Yon sirt	Боковая поверхность
Net of a cylinder	Silindr yoymasi	Развёртка цилиндра
Axial section	O'qli kesim	Осевое сечение
Hollow cylinder (pipe)	Bo'sh silindr	Полый цилиндр
Capacity	Sig'im	Вместимость
Solid of revolution	Aylanish jismi	Тело вращения

Stress the words in bold in the explanation above; ask for the Uzbek and Russian back before moving on.

05 Quick check**Check yourself**

The volume of a cylinder is:

$$2\pi rh$$

$$\pi r^2 h$$

$$\pi rh$$

$$\frac{1}{3} \pi r^2 h$$

The curved surface area is:

$$\pi r^2$$

$$2\pi rh$$

$$2\pi r^2$$

$$\pi r^2 h$$

The total surface area is:

$$2\pi rh$$

$$2\pi r(r + h)$$

$$\pi r(r + h)$$

$$\pi r^2 h$$

A pipe's cross-section area is:

$$\pi(R - r)^2$$

$$\pi(R^2 - r^2)$$

$$\pi R^2$$

$$2\pi R$$

A cylinder with a square axial section has:

$$h = r$$

$$h = 2r$$

$$h = 4r$$

$$r = 2h$$

06 Practice · 21 problems

Seven at each level. Set the easy row for everyone, the medium row for the main body of the class, and the hard row for those who finish early.

Easy · warm the idea up

7 PROBLEMS

1. Volume, $r = 3$, $h = 10$

ANSWER $90\pi \approx 283$

2. Volume, $r = 2$, $h = 7$

ANSWER $28\pi \approx 88$

3. Curved area, $r = 5$, $h = 4$

ANSWER 40π

4. Total area, $r = 1$, $h = 3$

ANSWER 8π

5. A tin of $r = 4$, $h = 10$: volume in litres

ANSWER ≈ 0.503

6. Volume, $r = 10$, $h = 10$

ANSWER 1000π

7. Curved area, $r = 7$, $h = 2$

ANSWER 28π

Medium · the standard question

7 PROBLEMS

1. $r = 5, h = 12$: volume and total area

ANSWER $300\pi; 170\pi$

2. A tin holds 500 ml and is 12 cm tall: its radius

ANSWER ≈ 3.64 cm

3. A cylinder of volume 1000π and radius 10: its height

ANSWER 10

4. Pipe, 2 m long, $R = 8, r = 6$: metal volume

ANSWER $5600\pi \approx 17\,600$ cm³

5. A cylinder with a square axial section and $r = 6$: its volume

ANSWER 432π

6. A cup (no lid), $r = 4, h = 9$: surface area

ANSWER 88π

7. A cylinder of total area 100π and $r = 5$: its height

ANSWER 5

Hard · stretch and reason

7 PROBLEMS

1. A closed cylinder of volume 1000 cm^3 with least surface area: find r

ANSWER $\approx 5.42 \text{ cm}$

2. A silo: cylinder $r = 3$, $h = 10$ with a hemisphere on top. Volume

ANSWER $108\pi \approx 339$

3. Same silo: external surface area (no base)

ANSWER 78π

4. A cylinder is inscribed in a cube of edge 10. Its volume

ANSWER 250π

5. A cylinder of height 15 has curved area 120π . Find r and V

ANSWER $r = 4$, $V = 240\pi$

6. A pipe of length L , outer R , wall thickness t : metal volume

ANSWER $\pi L(2Rt - t^2)$

7. Two similar cylinders of volumes 27 and 64: the area ratio

ANSWER $9 : 16$

07 Homework

Homework — 5 tasks

Say whether the solid is closed, open or hollow before choosing the formula.

1. A cylinder has radius 6 cm and height 15 cm. Find its volume and total surface area.
2. A cylindrical can holds 750 ml and has radius 5 cm. Find its height.
3. A pipe is 3 m long with outer radius 10 cm and inner radius 8 cm. Find the volume of metal in cm^3 .
4. A cylinder has a square axial section and radius 7 cm. Find its volume and total surface area.

5. A silo is a cylinder of radius 4 m and height 12 m with a hemispherical top. Find its capacity in m^3 .